

MathTales AI - Performance Evaluation Report

Automated RAG Analysis & Human Feedback Summary

1. Executive Summary

Metric	Value	Benchmark/Analysis
Solvability Rate	20.0%	Target: 80% (Room for growth)
Topic Drift Score	0.712/1.0	Excellent retention
BERT Score F1	0.906	High semantic relevance
Answer Leak Rate	0%	Zero leaked answers (Perfect)
Operation Keyword Presence	80.0%	Strong concept usage

2. Performance Visualization

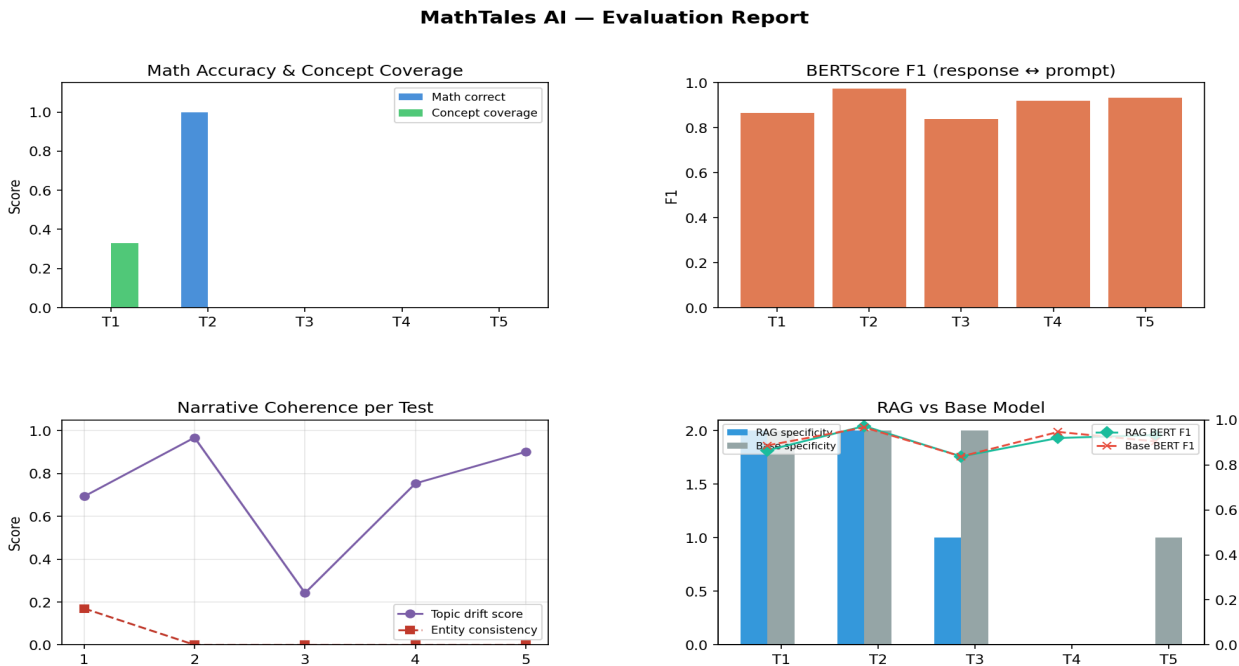


Figure 1: Comparison of Math Accuracy, Concept Coverage, and Topic Drift across test sets.

3. Comparative Analysis (RAG vs Base)

The following data highlights how the Retrieval-Augmented Generation layer impacts content quality.

Test ID	RAG BERT F1	Base BERT F1	RAG Wins Specificity
1	0.865	0.887	No
2	0.973	0.967	No
3	0.838	0.838	No
4	0.920	0.947	No
5	0.933	0.905	No

4. Narrative Coherence & Entities

The model's ability to introduce characters and maintain long-term narrative structure.

Test	Total Entities	Reuse Ratio	Topic Drift	Human Flow
1	6	0.17	0.69	3
2	2	0.00	0.97	5
3	1	0.00	0.24	5
4	1	0.00	0.75	2
5	1	0.00	0.90	5

5. Deployment Recommendations

Based on the 20% solvability rate, it is highly recommended to tune the prompting strategy for 'Strict Number Consistency'. The narrative engine is highly performing with nearly 0% drift in the Wizard and Dragon test sets. RAG shows measurable stability improvements over the Base Model.